Calibration

SWAT+ Documentation

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SWAT+ can change model parameters from inside the model, using two kinds of calibration data.

Hard data are measured time series at a point, such as daily streamflow at a gauge, sediment or nutrient loads, or groundwater levels. Fitting the model to hard data is done by an external tool (SWAT+ Toolbox, SWAT-CUP, a script) that writes a parameter change list to `calibration.cal`, runs SWAT+, compares the output with the observations and repeats. SWAT+ applies the change list at start-up, before the simulation begins.

Soft data are estimates of individual processes that are not measured at the site, or only as long-term averages: the share of precipitation that leaves as surface runoff or evapotranspiration, the baseflow share of streamflow, or an average county crop yield (Seibert and McDonnell, 2002; Arnold et al., 2015). SWAT+ fits these itself. After the normal run it adjusts a few water balance and plant parameters one at a time, rerunning the simulation after each change, until the average annual values for each land-use group match the targets.

The usual order is soft calibration first, so that each part of the water balance is about right, then hard calibration against the gauge record, which should then need mostly adjustments to peaks and recessions.

Pages

- Hard calibration. How SWAT+ applies a parameter change list, change types and object types.
- Parameter change files. Layout of `cal_parms.cal` and `calibration.cal`, and every parameter name SWAT+ accepts.
- Conditions and regionalization. Applying a change only to objects that match a condition, such as a soil group, land use or calibration group.
- Soft calibration. The `codes.sft` switches, the targets in each `.sft` file, the adjustment steps and the output files.
- Sensitivity workflow. Running a sensitivity analysis with an external sampler that writes `calibration.cal`.

References

Arnold, J. G., Youssef, M. A., Yen, H., White, M. J., Sheshukov, A. Y., Sadeghi, A. M., Moriasi, D. N., Steiner, J. L., Amatya, D. M., Skaggs, R. W., Haney, E. B., Jeong, J., Arabi, M., and Gowda, P. H. (2015). Hydrological processes and model representation: impact of soft data on calibration. Transactions of the ASABE, 58(6), 1637–1660.

Seibert, J., and McDonnell, J. J. (2002). On the dialog between experimentalist and modeler in catchment hydrology: use of soft data for multicriteria model calibration. Water Resources Research, 38(11), 1241.

Sources

- Calibration start-up: `src/proc_cal.f90`, called from `src/main.f90.in` before the simulation.
- Hard calibration changes: `src/cal_conditions.f90` and `src/cal_parm_select.f90`.
- Soft calibration: `src/calsoft_control.f90`, run from `main.f90.in` after the base simulation when `cal_soft = "y"`.
- Data types: `src/calibration_data_module.f90`.

Purpose

Soft calibration adjusts a small set of HRU and plant parameters until the simulated average annual water balance and crop yields match targets given for each land-use group. The targets are soft data: regional estimates such as the fraction of precipitation that leaves as surface runoff or evapotranspiration, the share of streamflow that is baseflow, or an average county yield. They are not time series.

Soft calibration runs inside the normal `swatplus` executable. After the base simulation ends, SWAT+ changes one parameter at a time and reruns the whole simulation period after each change. Each rerun is labelled on screen (`first esco adj`, `cn3_swf adj`, `epco adj`, and so on). A water balance calibration adds up to 17 full simulations in mode `a` and 20 in mode `b`, and plant calibration up to 13 more. The calibrated values are written to new files (see Output files); the model inputs are left unchanged.

In 62.0.1 two components work: the HRU water balance (`hyd` switch) and crop yield (`plnt` switch). The other switches are read, but they do not adjust anything. The status column of the switch table gives the detail for each.

Run sequence

1. At start-up SWAT+ reads the seven soft calibration files. A missing file, or a `null` name in `file.cio`, gives an empty table and that part is skipped. Without `codes.sft` soft calibration is off.
2. Any changes in `calibration.cal` are applied (see Hard calibration), so soft calibration starts from the changed values.
3. The base simulation runs as usual. At the end of each simulated year every HRU's annual water balance is added to its land-use group, and each harvest adds its yield to the plant targets.
4. If any switch in `codes.sft` is on, soft calibration runs next. It first sets the `print.prt` codes back to their defaults, which are mostly off, and turns on the average annual basin water balance. The warm-up period

(`nyskip`) is kept. Most output files therefore contain the base run, and `basin_wb_aa.txt` gains a record for each calibration run.
5. Each adjustment step changes one parameter for every HRU in each land-use group that is outside its tolerance, restores the starting state of the model, and reruns the full period. The steps are listed under Adjustment steps.
6. The calibrated parameters are written out.

Input files

The `chg` line of `file.cio` names nine files. The first two are the hard calibration files described in Parameter change files. The last seven belong to soft calibration:

Position on `chg` line	Default name	Earlier name	Content
3	`codes.sft`	`codes.cal`	component switches
4	`wb_parms.sft`	`ls_parms.cal`	limits for the water balance parameters
5	`water_balance.sft`	`ls_regions.cal`	water balance targets by land-use group
6	`ch_sed_budget.sft`	`ch_orders.cal`	channel erosion targets
7	`ch_sed_parms.sft`	`ch_parms.cal`	limits for the channel parameters
8	`plant_parms.sft`	`pl_parms.cal`	plant parameter start values and limits
9	`plant_gro.sft`	`pl_regions.cal`	crop yield targets

A file is only read when its name is on the `chg` line. A project written by SWAT+ Editor 4.0.1 can contain the `.sft` files while their names on the `chg` line are `null`. In that case edits to the files have no effect.

All files are free format. Values are separated by blanks, and names must not contain blanks or slashes. The first line of every file is a title and is skipped.

codes.sft

- Line 1: title.
- Line 2: header.
- Line 3: eight one-letter switches in the order below.

#	Column	Values	Status in 62.0.1
1	`hyd`	`n`, `a`, `b`, `y`	Works. `a` and `b` select the mode; `y` behaves as `b`.
2	`landscape`	`n`, `y`	Not usable. It is meant for simplified (`hru-lte`) HRUs, but its target table is never set up.
3	`plnt`	`n`, `y`	Works, crop yield targets only.
4	`sed`	`n`, `y`	Not usable, see Upland sediment.
5	`nut`	`n`, `y`	Nothing is calibrated. Setting `y` starts the soft calibration runs and changes nothing.
6	`ch_sed`	`n`, `y`	Channel results are summed, but no parameter is adjusted.
7	`ch_nut`	`n`, `y`	Nothing is calibrated.
8	`res`	`n`, `y`	Nothing is calibrated.

Soft calibration runs when `hyd` is anything other than `n` or any other switch is `y`.

Mode `a` calibrates each water balance component against a separate ratio of precipitation. Mode `b` uses only the water yield ratio and the baseflow share (`wyr` and `bfr` in `water_balance.sft`) and derives the other targets from them. The steps for each mode are under Adjustment steps.

Example (SWAT+ Editor 4.0.1 layout with `hyd` set to `b`):

```
codes.sft: written by SWAT+ editor v4.0.1 on 2026-09-13 21:18 for SWAT+ rev. 62
hyd           landscape     plnt          sed           nut           ch_sed        ch_nut        res
b             n             n             n             n             n             n             n
```

water_balance.sft

- Line 1: title.
- Line 2: number of regions.
- Line 3: header.
- For each region: a line with the region name and the number of land-use rows. If that number is above zero, a header line follows, then one target row per land-use group.

Use a single region. The HRU list of a region is filled only when the region is named `basin` or the file has one region, and in both cases the region contains every HRU. A second region gets no HRUs and is not calibrated. HRU areas are taken from `ls_unit.ele` (the `bsn_frac` of element i is used for HRU i).

Target rows:

#	Column	Units	Description
1	`name`	none	Land-use group. Matched to the `cal_group` column of `landuse.lum`. The name `basin` matches every HRU.
2	`surq_rto`	fraction of precipitation	Surface runoff / precipitation. Mode `a` target for `cn3_swf`. In mode `b` the value is replaced when the file is read by `wyr` × (1 − `bfr`).
3	`latq_rto`	fraction of precipitation	Lateral flow / precipitation. Mode `a` target for `latq_co`. In mode `b` it is replaced by 0.1 × `wyr` × `bfr`.
4	`perc_rto`	fraction of precipitation	Percolation below the soil profile / precipitation. Mode `a` target for `perco`. In mode `b` it is added to `wyr` to form the `esco` target.
5	`et_rto`	fraction of precipitation	Actual ET / precipitation. Mode `a` target for `esco` and `pet_co`. Not used in mode `b`.
6	`tileq_rto`	fraction	Read; not used as a target.
7	`pet`	mm	Read; not used.
8	`sed`	t/ha	Upland sediment yield target for the `sed` switch.
9	`wyr`	fraction of precipitation	Water yield / precipitation. Mode `b` target for `pet_co` and `esco`.
10	`bfr`	fraction of water yield	Share of water yield that is baseflow. Mode `b` only. Converted when the file is read to baseflow / precipitation (`wyr` × `bfr`), the target for `perco`.
11	`solp`	kg/ha	Read; not used.

The simulated values are built from the HRU annual water balance (the columns of `hru_wb_yr`), weighted by HRU area:

- precipitation: `precip`
- surface runoff: `surq_gen`
- lateral flow: `latq`
- percolation: `perc`

- ET: `et`
- water yield: `surq_gen + latq + perc + qtile`
- baseflow: `latq + perc + qtile`

An HRU is left out of these sums once it has received more than 10 mm of irrigation in total. Its parameters are still changed with the rest of its group. The ratios in a row are not checked to add up to 1.

Example written by SWAT+ Editor 4.0.1 for revision 62. The row contains mode `a` ratios. `wyr` and `bfr` are zero, so in mode `b` every target would be zero and nothing would be calibrated.

```
water_balance.sft: written by SWAT+ editor v4.0.1 on 2026-09-13 21:18 for SWAT+ rev. 62
1
name                    num
basin                     1
    name                surq_rto      latq_rto      perc_rto        et_rto     tileq_rto           pet
    basin                0.20000       0.10000       0.10000       0.50000       0.00000       0.90000
```

wb_parms.sft

- Line 1: title.
- Line 2: number of rows.
- Line 3: header.
- Then one row per parameter: `name chg_typ neg pos lo up`.

SWAT+ uses the rows by position and does not look at `name` or `chg_typ`. Keep the ten rows in the order below; renaming or reordering them changes which limits each step uses.

Row	`name`	Parameter changed	Used by	How it is changed
1	`cn2`	curve number of simplified HRUs	`landscape` switch; `pos` is also used by the `sed` switch	added
2	`esco`	`esco` in `hydrology.hyd`	modes `a` and `b`	added
3	`latq_co`	`latq_co` in `hydrology.hyd`	modes `a` and `b`	added
4	`petco`	`pet_co` in `hydrology.hyd`	modes `a` and `b`	multiplied
5	`slope`	HRU slope	`sed` switch	subtracted
6	`tconc`	time of concentration	`sed` switch	divided
7	`etco`	ET coefficient of simplified HRUs	`landscape` switch	added
8	`perco`	`perco` in `hydrology.hyd`	modes `a` and `b`	added
9	`revapc`	revap coefficient of simplified HRUs	`landscape` switch	added
10	`cn3_swf`	`cn3_swf` in `hydrology.hyd`, with the curve numbers recomputed	modes `a` and `b`	added

For the HRU water balance rows (2, 3, 4, 8, 10):

- `neg` and `pos` limit the adjustment made for one land-use group. For `petco` they limit the multiplier (Editor default 0.54 to 1.85). A group that reaches a limit is flagged, and some later runs skip it.
- `lo` and `up` bound the HRU parameter value after every change.

Mode `b` narrows the `petco` range for its first PET step. It halves `neg` and `pos` and moves `lo` and `up` halfway toward 1, then restores the full range for the second PET step. See Limits in 62.0.1 for what this does with the Editor defaults.

`cn3_swf` and `perco` are not changed on tile-drained HRUs. At start-up those HRUs are set to `cn3_swf` = 0.95 and `perco` = 0.1.

Example written by SWAT+ Editor 4.0.1 for revision 62:

```
wb_parms.sft: written by SWAT+ editor v4.0.1 on 2026-09-13 21:18 for SWAT+ rev. 62
10
name                      chg_typ          neg          pos           lo           up
cn2                        abschg     -6.00000      6.00000     35.00000     95.00000
esco                       abschg     -1.00000      1.00000      0.00000      1.00000
latq_co                    abschg     -1.00000      1.00000      0.00000      1.00000
petco                      abschg      0.54000      1.85000      0.70000      1.20000
slope                      pctchg    -25.00000     25.00000      0.00010      0.90000
tconc                      pctchg    -30.00000     30.00000      5.00000    960.00000
etco                       abschg     -0.40000      0.40000      0.80000      1.20000
perco                      abschg     -0.70000      0.70000      0.00100      1.00000
revapc                     abschg     -0.40000      0.40000      0.00000      0.40000
cn3_swf                    abschg     -1.20000      1.20000      0.00000      1.00000
```

plant_gro.sft

- Line 1: title.
- Line 2: number of regions.
- Line 3: header.
- For each region:
  - a line with `name`, the number of plants (`lum_num`), `nspu`, and then `nspu` HRU numbers;
  - if `lum_num` is above zero, a header line and one target row per plant.

With `nspu` = 0 the region covers every HRU. Otherwise `nspu` numbers follow, and a negative number closes a range: with `nspu` = 3, the list `1 -20 35` means HRUs 1 to 20 and HRU 35. Each HRU belongs to one plant region. `plant_parms.sft` is read second, so its region assignment is the one that applies.

Target rows:

#	Column	Units	Description
1	`name`	none	Plant name. Matched to the plant named in the harvest operation (`harv` or `hvkl` in a management schedule, `harvest` or `harvest_kill` in a decision table).
2	`yield`	t/ha	Target yield. Compared with the total harvested yield divided by the total harvested area over the whole run.
3	`npp`	t/ha	Read; not used.
4	`lai_mx`	none	Read; not used.
5	`wstress`	none	Read; not used.
6	`astress`	none	Read; not used.
7	`tstress`	none	Read; not used.

plant_parms.sft

- Line 1: title.
- Line 2: number of regions. Region i here is paired with region i of `plant_gro.sft`.
- Line 3: header.
- For each region:
  - a line with `name`, `lum_num`, `parms` and `nspu`;
  - if `lum_num` is above zero, a header line and `lum_num` × `parms` parameter rows.

Set `nspu` to 0 here, which puts every HRU in the region. With `nspu` above zero the region line is read a second time without the `parms` field, so the HRU list is misread.

Set `parms` to 4. The rows form four blocks of `lum_num` rows each, in this order: `pest_stress`, `epco`, `lai_pot`, `harv_idx`. Within every block the plants must be in the same order as in the matching `plant_gro.sft` region.

SWAT+ finds a row by its position in the block and always runs all four steps.

Parameter rows:

#	Column	Description
1	`var`	`pest_stress`, `epco`, `lai_pot` or `harv_idx`
2	`name`	plant name
3	`init_val`	start value, applied when the file is read to every plant of that name in the region's HRUs. This happens whether or not `plnt` is on.
4	`chg_typ`	read; not used
5	`neg`	lower limit of the adjustment. For `epco` it is the lowest `epco` value the calibration may set.
6	`pos`	upper limit of the adjustment. For `epco` it is the highest `epco` value the calibration may set.
7	`lo`	lower bound on the plant's value
8	`up`	upper bound on the plant's value

ch_sed_budget.sft and ch_sed_parms.sft

These two files are read, but no channel parameter is adjusted.

`ch_sed_budget.sft` layout:

- Line 1: title.
- Line 2: number of regions.
- Line 3: header.
- For each region: `name`, the number of target rows (`ord_num`), `nspu` and `nspu` channel numbers (0 means all channels). A header line always follows, then `ord_num` rows of `name chw chd hc fpd`.

The row fields are channel widening (mm/yr), downcutting or accretion (mm/yr), headcut advance (m/yr) and floodplain accretion (mm/yr).

`ch_sed_parms.sft` has the same layout as `wb_parms.sft`. Only `name` and `chg_typ` of its first row are used, and only when writing the channel summary.

The channel results are only summed when `ch_sed` is the only switch on, and only target rows named `basin` collect them.

Adjustment steps

In every step:

- The tolerance is the relative difference |target − simulated| / target, taken over the average annual values of the land-use group.
- A step only reruns the simulation if at least one group is outside tolerance.
- The first run of a step moves the parameter by a fixed rate. Later runs use a secant estimate from the last two runs.

Water balance, mode a

Step	Parameter	Target compared	Tolerance	Runs	First-run change	`wb_parms.sft` row
1	`esco`	`et_rto` × precipitation vs simulated ET	1 %	1 + 2	+ (target − simulated, mm) / 500	2
2	`pet_co`	same as step 1	1 %	2	multiplied by a ratio near 1	4
3	`cn3_swf`	`surq_rto` × precipitation vs `surq_gen`	2 %	1 + 2	− (target − simulated, mm) / 100	10
4	`latq_co`	`latq_rto` × precipitation vs `latq`	10 % first run, 5 % after	1 + 1	+ 0.0025 × (target − simulated, mm)	3
5	`perco`	`perc_rto` × precipitation vs `perc`	5 %	1 + 3	+ 0.001 × (target − simulated, mm)	8
6	`cn3_swf`	as step 3	2 %	1 + 2	as step 3	10

For example, with `surq_rto` = 0.2, 800 mm of precipitation and 120 mm of simulated surface runoff, the target is 160 mm, the difference is 40 mm, and the first `cn3_swf` run lowers `cn3_swf` by 0.4.

Water balance, mode b

Step	Parameter	Target compared	Tolerance	Runs
1	`pet_co`, half range	`wyr` × precipitation vs simulated water yield	1 %	2
2	`esco`	(`wyr` + `perc_rto`) × precipitation vs simulated water yield	1 %	1 + 2
3	`pet_co`, full range	as step 1	1 %	2
4	`cn3_swf`	`wyr` × (1 − `bfr`) × precipitation vs `surq_gen`	2 %	1 + 2
5	`latq_co`	0.1 × `wyr` × `bfr` × precipitation vs `latq`	10 % first run, 5 % after	1 + 2
6	`perco`	`wyr` × `bfr` × precipitation vs `latq` + `perc` + `qtile`	5 %	1 + 3
7	`cn3_swf`	as step 4	2 %	1 + 2

Crop yield

Each plant's average harvested yield is compared with its `plant_gro.sft` target:

1. Aeration stress. If the yield is more than 20 % below target, HRUs growing that plant that have no tile drain and an accumulated aeration stress above 50 get `perco` = 0.95. One run.
2. `epco`, with a 3 % tolerance. The first run sets `epco` to 1, or to a low value when the yield is more than 10 % above target. Four more runs narrow the interval between 0 and 1. The value is kept within `neg` and `pos` of the `epco` block.
3. `pest_stress`, changed by the relative yield error. One run.
4. `lai_pot`, 1 LAI unit for every 20 % of yield difference, then secant. 1 + 2 runs.
5. `harv_idx`, half the relative yield difference, then secant. 1 + 2 runs.

Upland sediment

The `sed` switch is meant to fit the `sed` targets by shortening the time of concentration and lowering the HRU slope, with the USLE slope-length factor recomputed. In 62.0.1 it does not work:

- Its loops run over the number of channel regions instead of the number of water balance regions.
- Simulated sediment is only summed when `hyd` is also on.
- The secant runs always compute a change of zero.

Treat the `sed` switch as unavailable.

Output files

File	Written when	Content
`hydrology-cal.hyd`	`hyd` or `plnt` on	One row per HRU with the calibrated `hydrology.hyd` values, written at the end.
`plant_parms.cal`	`plnt` on	Same layout as `plant_parms.sft`, with `init_val` updated to the calibrated values and `nspu` set to 0.
`hru-out.cal`	any switch on	Header only, except with the `landscape` switch, which appends the simplified HRU parameters.
`hru-new.cal`	`hyd` on	Opened but left empty.

To use the results in later runs, point file.cio at the new files:

- Use plant_parms.cal as plant_parms.sft. Its init_val values are applied when the file is read, also with plnt = n.
- Use hydrology-cal.hyd as hydrology.hyd, after the edits in the warning below. Rows are written per HRU under the name of its hydrology.hyd record, so every HRU needs a separate record. The extra 16th column (perco_lim, labelled NOT_USED) is ignored when the file is read.

  Converting hydrology-cal.hyd back to an input

  Two edits are needed before hydrology-cal.hyd can replace hydrology.hyd:

  - LAT_TTIME contains the value after the start-up conversion, 1 − exp(−1 / days). Read back as an input it is taken as days and converted a second time, which makes lateral flow drain much faster. Put the original lat_ttime values back before using the file.
  - When hyd and plnt are both on, the file contains the title, header and HRU rows twice. Keep only the second set, which is written after plant calibration.

Limits in 62.0.1

- Only one water balance region works (see water_balance.sft).
- wb_parms.sft and plant_parms.sft rows are found by position, so the row order is part of the format.
- In mode b the first PET step halves neg and pos for petco. With the Editor defaults (0.54, 1.85) the upper limit becomes 0.925. When simulated water yield is too high the step should raise PET; it caps the multiplier at 0.925 and lowers PET instead.
- plant_parms.sft must use nspu = 0 and parms = 4.
- The landscape, sed, nut, ch_sed, ch_nut and res switches do not calibrate anything (see codes.sft).
- Annual sums include the warm-up years, but the averages divide by the printed years only. Target ratios are unaffected because precipitation is treated the same way. The fixed first-run rates, which are in mm, act on values scaled by the same factor.

Related pages

- Hard calibration. Changes from calibration.cal, applied before the base run.
- Parameter change files. cal_parms.cal and calibration.cal, the first two files on the chg line.
- landuse.lum. The cal_group column that water_balance.sft rows are matched against.
- hydrology.hyd. The parameters written to hydrology-cal.hyd.

  Sources

  - Readers, called from src/proc_cal.f90 in this order: pl_read_regions_cal.f90 (plant_gro.sft), pl_read_parms_cal.f90 (plant_parms.sft), calsoft_read_codes.f90 (codes.sft), lcu_read_softcal.f90 (water_balance.sft), ls_read_parms_cal.f90 (subroutine ls_read_lsparms_cal, wb_parms.sft), ch_read_orders_cal.f90 (ch_sed_budget.sft), ch_read_parms_cal.f90 (ch_sed_parms.sft).
  - Control: src/calsoft_control.f90, called from src/main.f90.in after the base simulation when cal_soft = "y".
  - Water balance: src/calsoft_hyd.f90 (mode a); src/calsoft_hyd_bfr.f90 and its steps calsoft_hyd_bfr_pet, _et, _surq, _latq, _perc (mode b).
  - Crop yield: src/calsoft_plant.f90, calsoft_plant_zero.f90, pl_write_parms_cal.f90. Yields are added to the targets at harvest in mgt_sched.f90 and actions.f90.
  - Upland sediment: src/calsoft_sed.f90. Simplified HRU water balance: caltsoft_hyd.f90.
  - Annual sums and averages: calsoft_sum_output.f90 (end of every simulated year, called from time_control.f90) and calsoft_ave_output.f90 (end of every run).
  - Tile-drained HRU settings: topohyd_init.f90. lat_ttime conversion: hydro_init.f90. Output files are opened in header_write.f90.
  - Types: src/calibration_data_module.f90. File names: type input_chg in src/input_file_module.f90.

File and column	Fortran variable
codes.sft hyd, landscape, plnt, sed, nut, ch_sed, ch_nut, res	cal_codes%hyd_hru, %hyd_hrul, %plt, %sed, %nut, %chsed, %chnut, %res
water_balance.sft target row	type soft_calib_ls_processes: name, srr, lfr, pcr, etr, tfr, pet, sed, wyr, bfr, solp
wb_parms.sft rows	ls_prms(1:10)
plant_gro.sft target row	type soft_calib_pl_processes
plant_parms.sft parameter row	type pl_parms_cal; row k of block b is prm(k + (b-1)*lum_num)
HRU land-use group	hru%lum_group_c, from landuse.lum cal_group
accumulated aeration stress	strsa

Output file	Unit
hydrology-cal.hyd	5001
plant_parms.cal	107
hru-out.cal	4999
hru-new.cal	5000

Purpose

Hard calibration fits the model to measured time series such as streamflow, sediment or nutrient loads at a gauge. SWAT+ does not search for parameter values itself. An external tool (SWAT+ Toolbox, SWAT-CUP, a script) picks a set of parameter changes, writes them to calibration.cal, runs SWAT+, compares the output with the observations, and repeats. Each run of SWAT+ applies the changes in calibration.cal once, at start-up, and then simulates the full period with the changed parameters.

The input files are never edited. A change exists only for the run that reads it, so the same TxtInOut folder can be reused for every trial.

Files

Two files, both named on the chg line of file.cio, drive hard calibration:

- cal_parms.cal lists every parameter that may be changed, the object type it belongs to, and the lowest and highest value it may take. SWAT+ Editor writes this file and it rarely needs editing.
- calibration.cal is the change list for one run. Each row names a parameter from cal_parms.cal, the type and size of the change, and optionally the objects to change and the conditions they must meet.

If either slot is null or the file is missing, no changes are applied. The column layouts and the full list of parameter names are on Parameter change files.

Run sequence

1. SWAT+ reads all input files and sets up every object (HRUs, soils, plants, channels, aquifers, reservoirs).
2. It reads cal_parms.cal and calibration.cal.
3. For each row of calibration.cal, in file order, it goes through the listed objects (all objects of that type when none are listed), checks the row's conditions for each object, and changes the parameter on every object that meets them.
4. The simulation runs once with the changed parameters.

Because the changes are applied after the inputs are read and before the first day is simulated, they act for the whole simulation period. The year and day columns of calibration.cal only limit changes to the precipitation and temperature records (see Climate changes).

Rows are applied in order, and a later row for the same parameter acts on the value left by the earlier row. Two pctchg rows of 10 for cn2 give a 21 % increase.

Change types

Type	New value	Example (current value 70, change 5)
`absval`	the change value replaces the current value	5
`abschg`	current value + change	75
`pctchg`	current value × (1 + change / 100)	73.5
`relchg`	current value × (1 + change)	420

After every change the result is limited to the range `abs_min` to `abs_max` given for that parameter in `cal_parms.cal`. Use one of the four types; any other text in the change type column gives an undefined value.

Object types

The `obj_typ` column of `cal_parms.cal` tells SWAT+ which objects a parameter belongs to. When a row of `calibration.cal` lists no objects, every object of that type is changed.

Type	Objects changed when none are listed	Example parameters
`hru`	every HRU	`cn2`, `esco`, `epco`, `perco`, `lat_ttime`, `canmx`
`sol`	every HRU, in the soil layers from `lyr1` to `lyr2`	`awc`, `k`, `bd`, `usle_k`
`plt`	see Plant parameters	`lai_pot`, `harv_idx`, `phu_mat`
`hlt`	every HRU of the simplified `hru-lte` type	`cn2_lte`, `awc_lte`
`aqu`	every aquifer	`alpha`, `revap_co`, `flo_min`
`sdc`	every channel	`chn`, `chk`, `bank_exp`
`swq`	every channel nutrient parameter set	`rs1`, `bc1`, `mumax`
`res`	every reservoir	`k_res`, `evrsv`, `nsed`
`rdt`	every reservoir decision table	`drawdown_days`, `withdraw_rate`
`bsn`	the basin (applied once)	`surlag`, `cdn`, `nperco`
`gwf`	every gwflow cell	`aquifer_K`, `aquifer_Sy`
`gwf_riv`	every channel connected to gwflow	`stream_K`, `stream_thk`
`gwf_sgl`	applied once	`stream_bed`, `reservoir_K`
`cli`	see Climate changes	`precip`, `temp`

Soil layers

For `sol` parameters, `lyr1` and `lyr2` in `calibration.cal` set the first and last soil layer to change. A `lyr1` of 0 starts at the top layer, and a `lyr2` of 0 or more than the number of layers ends at the bottom layer.

Reservoir decision tables

For `rdt` parameters (`drawdown_days`, `withdraw_rate`) the five integer columns after `conds` select which storage zones of the decision table to change. A non-zero value in the column switches that zone on:

Column	Zone
`lyr1`	exclusive flood control storage
`lyr2`	seasonal flood control storage, non-flood season
`year1`	seasonal flood control storage, flood season
`year2`	multiple use storage, non-flood season
`day1`	multiple use storage, second zone

Plant parameters

Plant parameters with object type `plt` are changed in the plant community of each listed HRU, for the plants named in the row's conditions. Only `phu_mat`, `lai_pot` and `harv_idx` are handled this way. Each row needs a condition that names the plant (see Conditions and regionalization); a row without one changes nothing.

Give the HRUs to change in the object list of the row. When the list is empty, SWAT+ goes through as many HRUs as there are plants in plants.plt, starting from HRU 1.

To change epco use object type hru. That changes the HRU value and sets the same value for every plant in the HRU's community. With object type plt, an epco row changes lai_pot instead.

The plant carbon parameters bio_e, hvsti, rdmx, t_opt and t_base are listed with object type plt in the SWAT+ Editor cal_parms.cal. In 62.0.1 rows for these parameters change nothing.

Climate changes

Precipitation and temperature records can be changed over a date window with the parameters precip and temp, giving object type cli in cal_parms.cal. The change is applied to every daily value from day day1 to day day2 of each year from year1 to year2 (calendar years, inside the simulation period).

List the records to change in the object list of the row. With no list, nothing is changed. List one record per row. When a row lists several records, the change is applied to every listed record once for each record in the list, so a pctchg of 10 on three records gives a 33 % increase on each. In 62.0.1 temp rows do not change the listed temperature record.

Unknown names

A row whose name is not in cal_parms.cal, or whose name SWAT+ does not recognise as a parameter, is skipped without a message. Check spelling against the parameter list if a change seems to have no effect.

Related pages

- Parameter change files. Column layout of cal_parms.cal and calibration.cal, and the parameter names.
- Conditions and regionalization. Limiting a change to objects that meet a condition.
- Sensitivity workflow. Driving many runs from an external sampler.
- Soft calibration. Fitting the average annual water balance and crop yields before hard calibration.

Sources

- Readers: src/cal_parm_read.f90 (cal_parms.cal) and src/cal_parmchg_read.f90 (calibration.cal), called from src/proc_cal.f90.
- Applying changes: src/cal_conditions.f90, which calls src/cal_parm_select.f90 for each object. The change operator is src/chg_par.f90.
- Order of calls: src/main.f90.in. proc_cal runs after proc_hru, proc_cha, proc_aqu and the reservoir and wetland setup, and before time_control.
- Types: type calibration_parameters and type update_parameters in src/calibration_data_module.f90.

Purpose

cal_parms.cal is the list of parameters that calibration may change: the name of each parameter, the type of object it belongs to, and the lowest and highest value it may take. calibration.cal is the change list for one run: each row picks a parameter from cal_parms.cal, says how to change it, and optionally limits the change to some objects or to objects that meet conditions. How SWAT+ applies the list is described on Hard calibration.

Both files are named on the chg line of file.cio, in the first two positions (defaults cal_parms.cal and calibration.cal). A null name or a missing file switches the change list off.

All values are free format and separated by blanks. Names must not contain blanks.

cal_parms.cal

- Line 1: title (skipped).
- Line 2: number of parameter rows.

- Line 3: column header (skipped).
- Then one row per parameter.

#	Column	Description
1	`name`	Parameter name, as listed under Parameter names.
2	`obj_typ`	Type of object the parameter belongs to: `hru`, `sol`, `plt`, `hlt`, `aqu`, `sdc`, `rte`, `swq`, `res`, `rdt`, `bsn`, `pst`, `gwf`, `gwf_riv`, `gwf_sgl` or `cli`. It decides which objects are changed when a row of `calibration.cal` lists none.
3	`abs_min`	Lowest value the parameter may take. Every change is limited to this value.
4	`abs_max`	Highest value the parameter may take. Every change is limited to this value.
5	`units`	Units, for information only.

SWAT+ Editor writes this file with suitable limits for each parameter, and it rarely needs editing. A narrower range for a calibration run belongs in the calibration tool; the range here is the physical limit.

Example from `refdata/Osu_1hru` in the SWAT+ repository:

```
cal_parms.cal: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
186
name                      obj_typ       abs_min       abs_max            units
cn2                           hru      35.00000      95.00000             null
cn3_swf                       hru       0.00000       1.00000             null
usle_p                        hru       0.00000       1.00000             null
ovn                           hru       0.01000      30.00000             null
elev                          hru       0.00000    5000.00000                m
slope                         hru       0.00010       0.90000              m/m
slope_len                     hru      10.00000     150.00000                m
lat_ttime                     hru       0.50000     180.00000             days
```

The count on line 2 must match the number of rows. Rows past the count are not read.

calibration.cal

- Line 1: title (skipped).
- Line 2: number of change rows.
- Line 3: column header (skipped).
- Then one row per change, each followed directly by its condition lines, if it has any.

Change rows

#	Column	Description
1	`name`	Parameter name. Must also be in `cal_parms.cal`.
2	`chg_typ`	`absval` (replace), `abschg` (add), `pctchg` (percent change) or `relchg` (relative change). See Change types.
3	`val`	Size of the change.
4	`conds`	Number of condition lines that follow this row. 0 for none.
5	`lyr1`	First soil layer to change, for `sol` parameters (0 = top layer). For reservoir decision tables, a zone switch.
6	`lyr2`	Last soil layer to change, for `sol` parameters (0 = bottom layer). For reservoir decision tables, a zone switch.
7	`year1`	First year of the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.
8	`year2`	Last year of the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.
9	`day1`	First day of the year in the window, for `precip` and `temp`. For reservoir decision tables, a zone switch.

#	Column	Description
10	`day2`	Last day of the year in the window, for `precip` and `temp`.
11	`obj_tot`	Number of entries in the object list that follows on the same line. 0 means every object of the parameter's type.
12…	object list	`obj_tot` object numbers.

The object numbers are the sequence numbers of the objects (the first HRU is 1, and so on). A negative number closes a range started by the number before it, so `obj_tot` = 3 with the list `1 -20 35` means objects 1 to 20 and object 35.

Condition lines

Each change row is followed by `conds` condition lines. A condition line has one of two forms:

- `range <var> <min> <max>`: the object's value of `<var>` must lie between `<min>` and `<max>`. Only `res_pvol` is recognised.
- `<var> <alt> <targ> <targc>`: a test against the text value `<targc>`. `<alt>` and `<targ>` must be present but are not used by the conditions that exist.

The variables and how they are tested are on Conditions and regionalization.

Example

`refdata/Osu_1hru/calibration.cal`:

```
  hru-new.cal developed from soft data calibration
    13
 NAME           CHG_TYP                  VAL   CONDS  LYR1  LYR2   YEAR1  YEAR2   DAY1   DAY2  OBJ_TOT
cn2         pctchg                 7.714000E+00  0          0          0          0          0          0
petco       pctchg                -5.273950E+01  0          0          0          0          0          0
latq_co     pctchg                -3.732030E+01  0          0          0          0          0          0
alpha       pctchg                 6.859000E+00  0          0          0          0          0          0
epco        pctchg                 6.534100E+01  0          0          0          0          0          0
awc         pctchg                -6.256550E+01  0          0          0          0          0          0
esco        pctchg                 3.305500E+01  0          0          0          0          0          0
perco       pctchg                 1.434000E+00  0          0          0          0          0          0
cn3_swf     pctchg                -7.710000E-02  0          0          0          0          0          0
canmx       pctchg                -6.680230E+01  0          0          0          0          0          0
lat_len     pctchg                -6.241200E+00  0          0          0          0          0          0
revap_co    pctchg                -5.854530E+01  0          0          0          0          0          0
surlag      pctchg                -2.125720E+01  0          0          0          0          0          0
```

Every row has no conditions and an empty object list, so each change applies to all objects of its type. The rows have one more zero than the header has columns; the reader stops after `obj_tot` when it is 0, so the extra value is ignored.

A row that changes `cn2` by +5 only on HRUs 1 to 20 and HRU 35, and only where the hydrologic soil group is `C`:

```
cn2         abschg      5.0    1   0   0   0   0   0   0   3   1  -20   35
hsg         =           0      C
```

Parameter names

The names SWAT+ 62.0.1 recognises are listed below with the object type to give in `cal_parms.cal` and the input the parameter comes from. A name that is in neither list is skipped without a message. Some recognised names have no effect at 62.0.1, or change a different value; the notes with each name say which.

HRU parameters (hru)

Name	Input	Notes
cn2	condition II curve number from cntable.lum	The curve numbers for the other moisture conditions are recomputed.
cn3_swf	cn3_swf, hydrology.hyd	Not changed on tile-drained HRUs.
perco	perco, hydrology.hyd	Not changed on tile-drained HRUs.
esco, epco	esco, epco, hydrology.hyd	epco is also set for every plant in the HRU.
petco	pet_co, hydrology.hyd	
latq_co, lat_sed	latq_co, lat_sed, hydrology.hyd	
lat_ttime	lat_ttime, hydrology.hyd	Use absval with a value in days. abschg, pctchg and relchg act on an internal coefficient and give wrong travel times.
canmx	can_max, hydrology.hyd	
biomix	bio_mix, hydrology.hyd	
erorgn, erorgp	orgn_enrich, orgp_enrich, hydrology.hyd	
lat_orgn, lat_orgp	lat_orgn, lat_orgp, hydrology.hyd	
slope, slope_len, lat_len, dis_stream	slp, slp_len, lat_len, dist_cha, topography.hyd	
elev	HRU elevation	
ovn	Manning's n for overland flow, ovn_table.lum	
usle_p	USLE support practice factor, cons_practice.lum	
field_len, field_wid, field_ang	len, wd, ang, field.fld	
snofall_tmp, snomelt_tmp, snomelt_max, snomelt_min, snomelt_lag	fall_tmp, melt_tmp, melt_max, melt_min, tmp_lag, snow.sno	
tile_dep, tile_dtime, tile_lag, tile_rad, tile_dist, tile_drain, tile_pump	dp, t_fc, lag, rad, dist, drain, pump, tiledrain.str	
tile_latk	lat_ksat, tiledrain.str	The result is written to the tile lag time (lag).
cmn_h, nperco_h, pperco_h, phoskd_h, psp_h, nperco_lchtile_h	HRU values of the basin nutrient parameters cmn, nperco, pperco, phoskd, psp and nperco_lchtile	Change one HRU's copy. The basin names below change the basin value.

Soil layer parameters (sol)

Changed in soil layers lyr1 to lyr2 of each HRU.

Name	Input	Notes
z	layer depth dp, soils.sol	
bd, awc, k	bd, awc, soil_k, soils.sol	Dependent water contents are recomputed after awc changes.
clay, silt, sand, rock	clay, silt, sand, rock, soils.sol	
alb, usle_k, ec, cal, ph	alb, usle_k, ec, caco3, ph, soils.sol	
anion_excl, crk	anion_excl, perc_crk, soils.sol	Whole profile values.
cbn	carbon, soils.sol	No effect: the soil carbon pools are set up after the change list is applied.
lab_p, hum_c_n, hum_c_p	lab_p, hum_c_n, hum_c_p, nutrients.sol	No effect, for the same reason.

Plant parameters (plt)

Changed for the plants named in the row's plant conditions, in the plant community of each listed HRU. See Plant parameters for how to write these rows.

Name	Input
lai_pot	lai_pot, plants.plt

Name	Input
harv_idx	harv_idx, plants.plt
phu_mat	heat units to maturity of the plant in the HRU's community, computed at start-up from the plant and the climate

bio_e, hvsti, rdmx, t_opt, t_base and usle_c are recognised names, but rows for them change nothing.

Basin parameters (bsn)

Changed once for the whole basin.

Name	Input
surlag, prf, evrch, percop, cdn, sdnco, tb_adj, n_updis, rsd_covco	parameters.bsn
init_seq, init_microb, init_slow, init_passive, koc_c, solc_ratio, manure_c_frac, bio_consol, till_consol, sfc_rsd_photodeg, n_act_frac, cnr_cap, cnr_ref, cpr_cap, cpr_ref, t_cbn_min, t_cbn_opt, t_cbn_max	carbon.bsn
bmix_a, bmix_b, bmix_c, tillmix_a, tillmix_b, tillmix_c	biological and tillage mixing curve coefficients, carbon.bsn

Basin rows for adj_pkr, ffcb, dorm_hr, cmn, nperco, pperco, phoskd, psp, nperco_lchtile, rsdco, msk_co1, msk_co2, msk_x, plaps, tlaps and evlai change nothing. SWAT+ copies these basin values to the HRUs, channels, plants and weather stations while it sets them up, before the change list is applied, and does not read the basin value again. For the nutrient parameters, use the HRU names (cmn_h, nperco_h, pperco_h, phoskd_h, psp_h, nperco_lchtile_h) instead.

The per-layer carbon rates of carbon_lyr.bsn (hp_rate, hs_rate, microb_rate, meta_rate, str_rate, microb_top_rate, hs_hp, a1co2, asco2, apco2, abco2) are recognised names, but rows for them change nothing.

Aquifer parameters (aqu)

Name	Input
flo_init_mm, dep_bot, dep_wt_init, no3_init, minp_init, cbn_init, flo_dist, bf_max, alpha, revap_co, deep_seep, sp_yld, hlife_n, flo_min, revap_min	gw_flo, dep_bot, dep_wt, no3_n, sol_p, carbon, flo_dist, bf_max, alpha_bf, revap, rchg_dp, spec_yld, hl_no3n, flo_min, revap_min, aquifer.aqu

Channel parameters (sdc or rte)

Name	Input
chw, chd, chs, chl, chn, chk, cov, d50, ch_clay, carbon, ch_bd, chss, bank_exp, vcr_coef, bankfull_flo, fps, fpn, hc_kh, hc_hgt, hc_ini, ch_n_conc, ch_p_conc, ch_p_bio, pk_rto, fp_inun_days, n_setl, p_setl, n_sol_part, p_sol_part, n_dep_enr, p_dep_enr, arc_len_fr, bed_exp, wash_bed_fr	channel geometry, hydraulics, sediment and nutrient parameters of the SWAT-DEG channel inputs

Channel nutrient parameters (swq)

Name	Input
mumax, rs1 to rs7, rk1 to rk6, bc1 to bc4	in-stream nutrient and algae rates, nutrients.cha (see channel inputs)

A rk5 row changes rs2 (the ben_disp column of nutrients.cha) instead.

Reservoir parameters (res)

Name	Input
esa, evol, psa, pvol	emergency and principal spillway surface areas and volumes, hydrology.res
k_res, evrsv	bottom hydraulic conductivity and evaporation coefficient, hydrology.res
nsed, res_d50, sed_stlr, velsetlr	normal sediment concentration, median particle size, settling rate and settling velocity, sediment.res
psetlr1, psetlr2, nsetlr1, nsetlr2, nsolr, psolr, soln_stl_fr, solp_stl_fr	nutrient settling rates, nutrients.res
vol, sed, orgn, orgp, no3, nh3, no2, solp	water, sediment and nutrient content at the start of the run, initial.res

Reservoir decision table parameters (rdt)

Name	Input
drawdown_days, withdraw_rate	release rules of the reservoir decision tables in res_rel.dtl, per storage zone (see Reservoir decision tables)

Pesticide parameters (pst)

List the pesticides to change (their order in pesticide.pes) in the object list. With an empty list nothing is changed.

Name	Input
pst_koc, pst_washoff, pst_foliar_hlife, pst_soil_hlife, pst_solub, pst_aq_hlife, pst_aq_volat, pst_aq_settle, pst_aq_resus, pst_ben_hlife, pst_ben_bury, pst_ben_act_dep	soil_ads, frac_wash, hl_foliage, hl_soil, solub, aq_hlife, aq_volat, aq_settle, aq_resus, ben_hlife, ben_bury, ben_act_dep, pesticide.pes

Simplified HRU parameters (hlt)

Name	Input
cn2_lte, awc_lte, etco_lte, tc_lte, soildep_lte, slope_lte, slopelen_lte, sy_lte, abf_lte, revapc_lte, percc_lte, sw_lte, gw_lte, gwflow_lte, gwdeep_lte, snow_lte, uslek_lte, uslec_lte, uslep_lte, uslels_lte	hru-lte.hru

awc_lte is given per mm of soil depth; SWAT+ multiplies the change and the upper limit by the soil depth.

gwflow parameters

These only act when gwflow is switched on in codes.bsn. Each name needs the object type shown.

Name	Object type	Objects changed	Input
aquifer_K, aquifer_Sy, aquifer_exdp	gwf	gwflow cells	aquifer hydraulic conductivity, specific yield and extinction depth, gwflow inputs

Name	Object type	Objects changed	Input
`tile_K`, `floodplain_K`	`gwf`	gwflow cells	tile drain and floodplain conductivity, only when tile or floodplain exchange is on
`aquifer_delay`	`hru`	HRUs	recharge delay from the soil to the water table, in days
`stream_K`, `stream_thk`	`gwf_riv`	channels	streambed conductivity and thickness of the cells connected to each channel
`stream_bed`	`gwf_sgl`	applied once	streambed elevation offset, one value for all channels
`reservoir_K`, `pond_bed_K`	`gwf_sgl`	applied once	reservoir and pond bed conductivity

Climate parameters (`cli`)

Name	Input
`precip`	daily precipitation of the listed `*.pcp` records, from `day1` to `day2` of each year from `year1` to `year2`
`temp`	daily temperature of the listed `*.tmp` records; changes nothing

See Climate changes before using these.

Related pages

- Hard calibration. When and how the change list is applied.
- Conditions and regionalization. Condition line variables.
- Soft calibration. The seven `.sft` files that follow these two on the `chg` line.

Sources

- Readers: `src/cal_parm_read.f90` and `src/cal_parmchg_read.f90`. Object list expansion: `src/define_unit_elements.f90`.
- Parameter names and the variables they change: `src/cal_parm_select.f90`. Object type dispatch: `src/cal_conditions.f90`.
- Types: `type calibration_parameters`, `type calibration_conditions` and `type update_parameters` in `src/calibration_data_module.f90`. File name slots: `in_chg%cal_parms` and `in_chg%cal_upd` in `src/input_file_module.f90`.

Column	Fortran variable
`cal_parms.cal` `name`, `obj_typ`, `abs_min`, `abs_max`, `units`	`cal_parms(i)%name`, `%ob_typ`, `%absmin`, `%absmax`, `%units`
`calibration.cal` `name` to `day2`	`cal_upd(i)%name`, `%chg_typ`, `%val`, `%conds`, `%lyr1`, `%lyr2`, `%year1`, `%year2`, `%day1`, `%day2`
`calibration.cal` `obj_tot` and list	`cal_upd(i)%num_tot`, expanded into `cal_upd(i)%num(:)`
condition `var alt targ targc`	`cal_upd(i)%cond(j)%var`, `%alt`, `%targ`, `%targc`
condition `range var min max`	`cal_upd(i)%cond(j)%var`, `cal_upd(i)%val1`, `cal_upd(i)%val2` (shared by all range lines of the row)

Purpose

A row of calibration.cal can be followed by condition lines. SWAT+ then changes the parameter only on the objects that meet every condition of the row. Conditions are checked once, at start-up, against the state of each object after the inputs are read.

Conditions and object lists are the two ways to calibrate parts of a watershed separately, for example one set of curve number changes for cropland and another for forest, or different groundwater parameters upstream and downstream of a gauge.

Condition lines

The number of condition lines is given in the **conds** column of the change row, and the lines follow the row directly. Each line has one of two forms.

Range condition

`range <variable> <min> <max>`

The object's value must be greater than **<min>** and less than **<max>**.

Text condition

`<variable> <operator> <number> <text>`

The object's value must equal **<text>**. All four values must be present. The operator and number are read but none of the existing conditions uses them, so any placeholder works (for example **=** and **0**).

When a row has several condition lines, an object must meet all of them.

Condition variables

Variable	Form	Objects	Condition met when
cal_group	text	HRU	the **cal_group** of the HRU's land use in landuse.lum equals the text
landuse	text	HRU	the HRU's land use name (the **lu_mgt** entry of hru-data.hru) equals the text
hsg	text	HRU	the hydrologic soil group of the HRU's soil (**hyd_grp** in soils.sol) equals the text, for example **C**
texture	text	HRU	the **texture** of the HRU's soil in soils.sol equals the text
plant	text	HRU	the first plant of the HRU's plant community is the named plant
pl_class	text		see below
res_typ	text	reservoir decision table	the decision table's name equals the text
res_pvol	range	reservoir	the principal spillway volume, in 10^4 m^3 (ha-m, as in hydrology.res), is between **<min>** and **<max>**

The HRU conditions apply to parameters whose object type is **hru**, **sol** or **plt**. They compare against the HRU with the same number as the object being changed, so with other object types (aquifers, channels) they test an unrelated HRU.

Text comparisons are exact and case sensitive, and an unknown variable name is ignored, so a misspelt condition lets every object through.

Notes on specific variables:

- **plant**: only the first plant in the community is checked. For a community of corn and soybean in that order, **plant = corn** matches and **plant = soyb** does not. For the plant parameters **lai_pot**, **harv_idx** and **phu_mat** the **plant** condition has a second role: it names the plant whose value is changed (see Plant parameters).

- `pl_class`: compares the class of the plant at the same position in `plants.plt` as the object number. It does not look at the HRU's plants, so it is not useful for selecting HRUs.
- `res_pvol`: both limits are taken from the last `range` line of the row.

Regional calibration

Calibration groups

The most direct way to calibrate regions or land-use groups separately is the `cal_group` column of `landuse.lum`:

1. Give each land use a group name in `cal_group`, for example `crop_north`, `crop_south`, `forest`. Land uses that share a name form one group.
2. Write one change row per group, each with a `cal_group` condition.

```
cn2      pctchg   -8.0   1   0  0  0  0  0  0   0
cal_group  =  0  crop_north
cn2      pctchg    4.0   1   0  0  0  0  0  0   0
cal_group  =  0  forest
```

The same `cal_group` names are used by soft calibration to match its water balance targets to HRUs.

Object lists

The other way is to list the objects in the row itself (`obj_tot` and the numbers after it), for example the HRUs, aquifers or channels upstream of a gauge. A negative number closes a range, so `3  1 -120 245` selects objects 1 to 120 and 245. Object lists work for every object type.

Object lists and conditions can be combined: the conditions are then checked only on the listed objects.

Scheduled land use updates

SWAT+ also reads a file named `scen_dtl.upd` from the project folder, if it exists. It lists decision tables from the scenario decision table file (`scen_lu.dtl` by default, named on the `decision_table` line of `file.cio`) that are checked for every HRU at the start of every simulated day. Their actions can change land use, management or other HRU settings during the run, for example to simulate land use change at a set date.

- Line 1: title.
- Line 2: number of tables.
- Line 3: header.
- One line per table: maximum number of times the table may act, table type, and table name.

The maximum and the type are read but not used in 62.0.1: every listed table is checked every day for every HRU.

This file is separate from `calibration.cal`. Its changes happen during the run, while `calibration.cal` changes happen before the run starts. See Decision tables for how tables are written.

Related pages

- Parameter change files. Layout of the change rows.
- Hard calibration. When and how changes are applied.
- landuse.lum. The `cal_group` column.

Sources

- Reading condition lines: `src/cal_parmchg_read.f90`.
- Evaluating conditions: `src/cal_conditions.f90`.
- Calibration group of an HRU: `src/hrudb_init.f90` copies `lum(ilu)%cal_group` to `hru(ihru)%cal_group`.
- `scen_dtl.upd`: `src/cal_cond_read.f90`; used in `src/time_control.f90`.
- Types: `type calibration_conditions`, `type update_parameters` and `type update_conditional` in `src/calibration_data_module.f90`.

Condition variable	Compared with
cal_group	hru(ielem)%cal_group
landuse	hru(ielem)%land_use_mgt_c
hsg	soil(ielem)%hydgrp
texture	soil(ielem)%texture
plant	pcom(ielem)%pl(1)
pl_class	pl_class(ielem)
res_typ	dtbl_res(ielem)%name
res_pvol	res_ob(ielem)%pvol, against 10000 * val1 and 10000 * val2

Purpose

A sensitivity analysis shows which parameters change a chosen model result the most, so that calibration can concentrate on them. SWAT+ does not include a sensitivity analysis. It is run by an external tool that writes `calibration.cal`, runs SWAT+, reads a result from the output files, and repeats for many parameter sets. Each run applies the changes in `calibration.cal` at start-up and leaves the input files untouched, so one project folder serves every run.

SWAT+ Toolbox does this from a graphical interface. The steps below are what any tool or script has to do.

Steps

1. Choose the parameters to test and a range for each. The names are those on Parameter names. The `abs_min` and `abs_max` values in `cal_parms.cal` are physical limits; a narrower range around reasonable values usually gives more useful results.
2. Generate the samples: use a sampling design that matches the analysis method: Sobol (Saltelli sampling), FAST, Morris (elementary effects), Latin hypercube with one-at-a-time changes (LH-OAT), and so on. Each sample is one set of parameter values. Libraries such as SALib (Python) or `sensitivity` (R) provide both the sampling and the analysis.
3. For each sample, write `calibration.cal` with one row per parameter, run SWAT+ in the project folder, and compute the result of interest: total or peak flow at an outlet, a performance measure against observations (NSE, KGE, percent bias), a sediment or nutrient load.
4. Compute the indices: pass the sample matrix and the vector of results to the analysis method: first-order and total-order indices for Sobol and FAST, mean and spread of elementary effects for Morris.

The runs are independent, so they can be spread over several processor cores, each in a separate copy of the project folder. Sample sizes of several hundred to a few thousand runs are common.

Writing calibration.cal

A sample with three parameters, written as relative and absolute changes:

```
sensitivity sample 17
3
NAME      CHG_TYP      VAL    CONDS  LYR1  LYR2  YEAR1  YEAR2  DAY1  DAY2  OBJ_TOT
cn2       pctchg     -6.2     0      0     0     0      0      0     0     0
esco      absval      0.71    0      0     0     0      0      0     0     0
alpha     absval      0.048   0      0     0     0      0      0     0     0
```

Points to watch:

- `pctchg` scales each object's current value, so HRUs keep their differences. `absval` sets every object to the same value.
- Every parameter name must also be in `cal_parms.cal`, and its limits there still apply. A sample value outside `abs_min` to `abs_max` is cut to the limit, which flattens the response near the edges.

- A misspelt name is skipped without a message. Check a first run by comparing two extreme samples before starting the full set.
- Some parameter names have no effect in 62.0.1, or work only with restrictions; see the notes on Parameter change files and Hard calibration before including them.

Reading results

Daily channel output is in `channel_sd_day.txt` (or `.csv`), with flow out of each channel in the `flo_out` column (m^3/s). Switch on only the output the analysis needs in `print.prt`; writing every file for thousands of runs costs a lot of time and disk space. The output files are described in the Output Reference.

Related pages

- Parameter change files. Format of `cal_parms.cal` and `calibration.cal`, and the parameter names.
- Hard calibration. How each run applies the change list.

Sources

- The change list is read by `src/cal_parmchg_read.f90` and applied by `src/cal_conditions.f90`, both from `src/proc_cal.f90` before the simulation starts.